Machine learning Development life cycle:- (MLDLC / MLDC) #10 day of 100 days

SDLC = Software development life cycle

Machine learning development life cycle
-> set of guidlince जो आपको follow करने होते हैं। अगर भी आप कोई machine learning based software product बनाते हो। Every-guidline.

Ex:- आपको कोई software product बनाना है जिसमे machine learning हो। (could be anything ex- loan prediction, recommendation system.

Step 1 -> frame the problem
[correctly frame the problem, ex:- how much cost, mode - offline / online]

Step 2 -> Gathering the data.
[like small think in collage project data जो easily get from kaggle or anywhere]
But for company you want very specific.

Data

- CSV
- API
- web scraping
- spark, cluster
- database → (datawarehouse (ETL))

3. Data pre processing :-

(Bahut sari problem hoon sakti hai data ~~n~~ noise, wagera)

- remove duplicate
- remove missing value
- remove outlier
- → scale value (like very small or big)

4. Exploratory Data analysis

→ data ko Analysis karte ~~e~~

→ Rel^n b/w Input and output

→ Viz analysis.

→ Univariate / Bivariate.

→ Inbalance →

5. feature Engineering and selection

→ Based on Input

ex:- house price

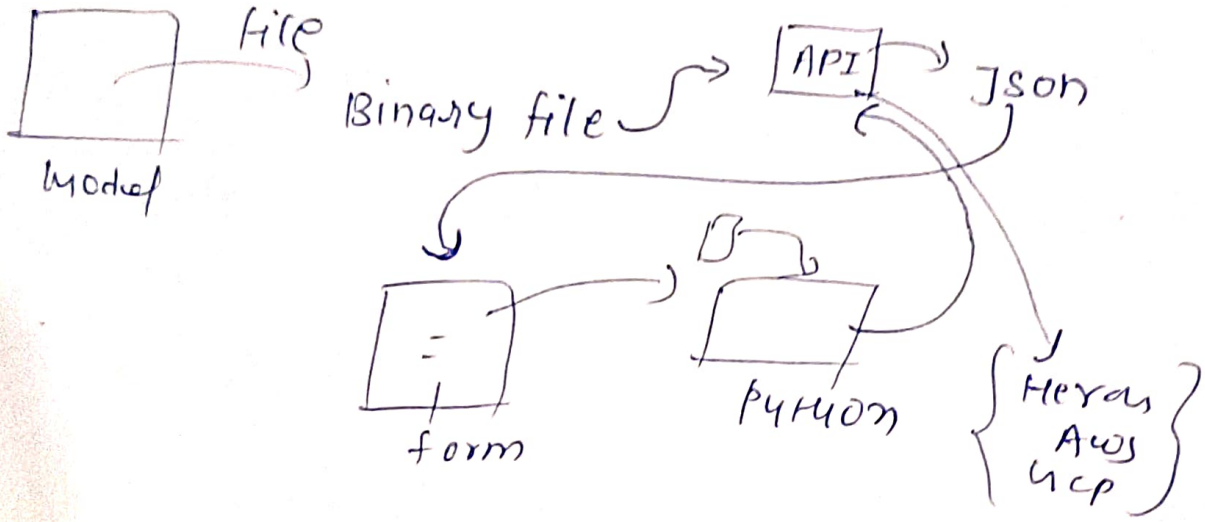
6. Model Training, Evaluation and selection.

⇒ diff.. diff algorithm and based on trend the data

→ evaluation → matrix (Accuracy square,)

7. Model deployment:-

↳ convert into software (APP, web)

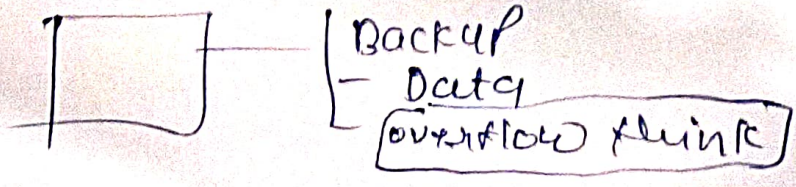


8. Testing

↳ loyal costume - rollout

Ex- like an software update होता
पहले धीरे - धीरे सभी को change
roll Based.

A/B Testing

9. Optimize.

Server पर model launch करते है
for all costumer